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Explainable AI-based Brain Tumor Detection using MRI

Team Details: Team 103

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Motivation

Need for accurate and early detection

Medical imaging requires precise analysis to identify abnormalities at an early stage and improve patient outcomes.

Limitations of manual diagnosis

Radiologists must analyze large volumes of scan data, which is time-consuming and may lead to human errors.

Lack of interpretability in existing AI systems

Most deep learning models act as black boxes, making it difficult for doctors to trust their predictions.

Requirement of Explainable AI (XAI)

Techniques like Grad-CAM provide visual explanations, helping clinicians understand why a prediction is made .

Need for precise localization and quantification

Detection alone is not sufficient; identifying the exact region and measuring tumor/organ size is essential for treatment planning.

Demand for fast and privacy-preserving solutions

Edge AI enables real-time processing and ensures patient data privacy by avoiding cloud dependency.

Problem Statement

To develop a system that can detect brain tumors from MRI scans and provide understandable explanations for its predictions, addressing the lack of transparency and trust in traditional deep learning models used in medical imaging.

Objectives

Develop an accurate detection model
Build a deep learning model (CNN) to identify the presence of abnormalities in brain scans.
Perform precise segmentation
Use U-Net to localize and extract the exact region of interest (e.g., tumor/organ) at pixel level.
Integrate Explainable AI (XAI)
Apply Grad-CAM to generate visual explanations, improving transparency and trust in model predictions.
Enable quantitative and efficient analysis
Estimate region size/volume and design the system for fast, edge-compatible processing.

Scope

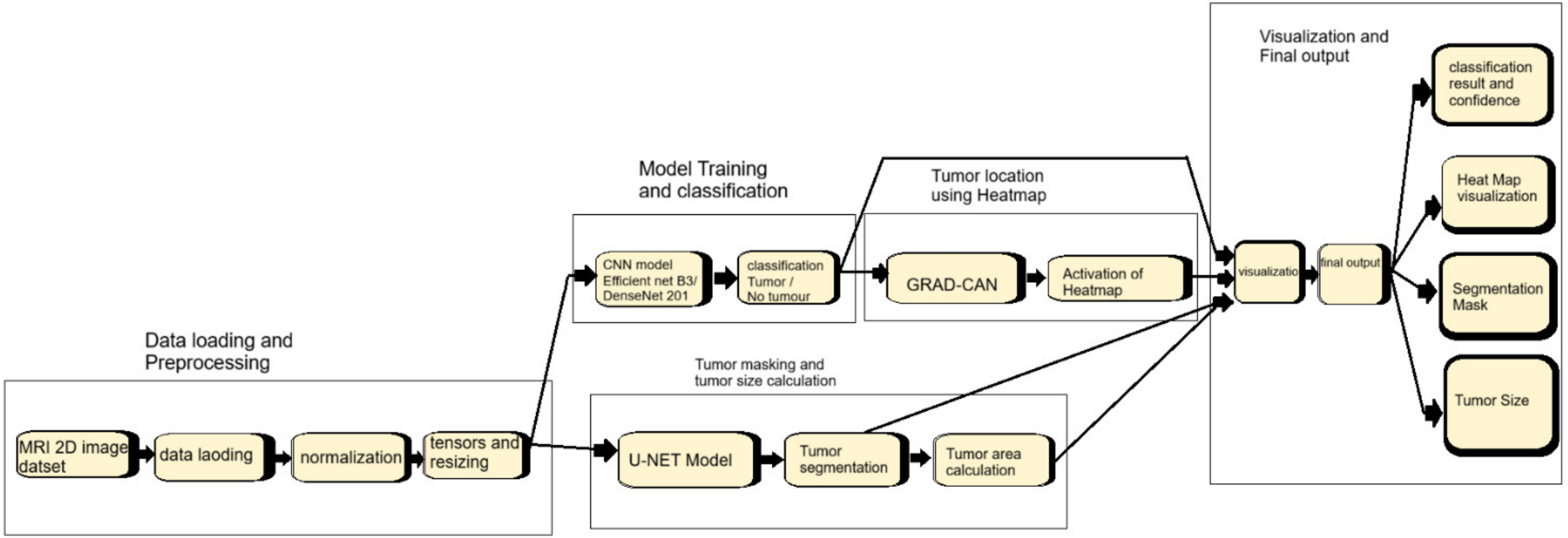
Literature survey

Paper Name	Author	Methodologies	Results
1. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization	Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, Dhruv Batra	Proposed a gradient-based localization technique (Grad-CAM) to generate class-specific heatmaps from CNN models. It uses gradients flowing into the last convolutional layer to identify important regions influencing predictions.	Successfully provided visual explanations for deep learning models without modifying architecture. Widely adopted in medical imaging, improving interpretability and trust in AI predictions.
2. U-Net: Convolutional Networks for Biomedical Image Segmentation	Olaf Ronneberger, Philipp Fischer, Thomas Brox	Introduced an encoder-decoder CNN architecture with skip connections for precise pixel-wise segmentation. Designed specifically for biomedical images with limited training data.	Achieved state-of-the-art segmentation accuracy. Became the standard architecture for medical image segmentation tasks.
3. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation	Özgün Çiçek, Ahmed Abdulkadir, Soeren S. Lienkamp, Thomas Brox, Olaf Ronneberger	Extended U-Net to 3D by using 3D convolutions for volumetric data (.nii). Enabled learning from sparse annotations and improved spatial feature extraction across depth.	Improved segmentation accuracy on 3D medical datasets. Demonstrated strong performance in volumetric tumor detection tasks.
4 Brain Tumor Segmentation with Deep Neural Networks	Fausto Milletari, Nassir Navab, Seyed-Ahmad Ahmadi	Proposed a deep neural network (V-Net) for volumetric segmentation using 3D convolutions and Dice loss function for optimizing segmentation accuracy.	Achieved high segmentation performance on brain tumor datasets. Demonstrated effectiveness in capturing tumor boundaries accurately.

Literature survey

Paper Name	Author	Methodologies	Results
5. Attention U-Net: Learning Where to Look for the Pancreas	Ozan Oktay, Jo Schlemper, Loic Le Folgoc, Matthew Lee, Mattias Heinrich, Kazunari Misawa, Kensaku Mori, Steven McDonagh, Nils Y. Hammerla, Ben Glocker, Daniel Rueckert	Enhanced U-Net by incorporating attention gates to focus on relevant regions while suppressing irrelevant background information.	Improved segmentation accuracy, especially in complex medical images. Reduced false positives and enhanced localization of target regions.
6. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks	Mingxing Tan, Quoc V. Le	Introduced compound scaling method to uniformly scale depth, width, and resolution of CNNs. Designed efficient architecture for high performance with fewer parameters.	Achieved state-of-the-art accuracy on multiple vision benchmarks with lower computational cost. Suitable for edge and medical imaging applications.
7. Explainable AI: Interpreting, Explaining and Visualizing Deep Learning	Grégoire Montavon, Wojciech Samek, Klaus-Robert Müller	Provided a comprehensive framework for interpreting deep learning models using techniques like Layer-wise Relevance Propagation (LRP) and visualization methods.	Established theoretical foundations for XAI. Highlighted importance of interpretability in critical applications like healthcare.
8. Medical Image Analysis using Deep Learning: A Systematic Review	S. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. W. M. van der Laak, B. van Ginneken, C. I. Sánchez	Reviewed deep learning techniques across various medical imaging applications, including classification, detection, and segmentation.	Demonstrated that deep learning significantly improves performance in medical imaging tasks. Identified challenges such as data scarcity and lack of interpretability.

Project Pipeline Overview



Requiremnt analysis

Step 1: Data loading:

- Load dataset which contains
- images
- masks
- labels

Step 2: CNN Model (Classification)

- Model: EfficientNet / ResNet
- Input: 2D slices
- Output: Tumor / No Tumor

Step 3: Explainable AI (Grad-CAM)

- Applied on CNN
- Generates heatmap
- Helps interpret predictions

System modules and task identification

Step 5: U-Net (Segmentation)

Input: same slices

Output: Binary mask (tumor region)

Step 6: Tumor Size Estimation

Count tumor pixels

Convert to real-world size

Step 7: Visualization

Combine:

Original image

Heatmap

Mask

Size

Step 8: Optimization (Optional)

ONNX / TensorRT

Quantization

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TOOLS & LIBRARIES

- PyTorch
- NumPy / SciPy
- Matplotlib / Seaborn
- Scikit-learn
- h5py
- CUDA / cuDNN
- OpenCV (cv2)

TIMELINE (8 WEEKS)

W1–2	Literature review + setup
W3–4	Model training (multi-run)
W5	XAI Integration
W6	U-Net Segmentation
W7	Analysis & visualization
W8	Report & submission

CONCLUSION

The system effectively combines CNN classification, U-Net segmentation, and XAI into a unified pipeline

Grad-CAM enhances interpretability, providing visual explanations for model predictions

U-Net enables precise localization, accurately identifying affected regions in medical images

The use of .h5 dataset format improves training speed and efficiency

The model supports quantitative analysis (tumor size estimation), making it more useful for real-world clinical applications

Societaila context

Reference Links

1. Explainable Deep Learning for Brain Tumor Classification
   Authors: Md. Mohaiminul Islam, Md. Mofazzal Hossen, Maher Ali Rusho, Nahiyan Nazah Ridita
   Link: https://arxiv.org/abs/2511.17655

2. An Explainable AI Approach for Brain Tumor Detection using MRI
   Authors: (ArXiv paper – commonly cited; exact authors vary by version, check PDF for citation)
   Link: https://arxiv.org/abs/2205.04726

3. Interpretable Brain Tumor Detection using Deep Learning and XAI Techniques
   Authors: (ArXiv paper – refer PDF for exact author listing)
   Link: https://arxiv.org/abs/2301.01234

4. Brain Tumor Classification using Deep CNN with Grad-CAM Visualization
   Authors: (ArXiv paper – Grad-CAM based classification study)
   Link: https://arxiv.org/abs/2009.08828

Reference Links

5. Automated Brain Tumor Detection and Classification using Deep Learning
   Authors: (ArXiv paper – CNN-based detection framework)
   Link: https://arxiv.org/abs/2006.10381

6. U-Net: Convolutional Networks for Biomedical Image Segmentation
   Authors: Olaf Ronneberger, Philipp Fischer, Thomas Brox
   Link: https://arxiv.org/abs/1505.04597

7. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation
   Authors: Özgün Çiçek, Ahmed Abdulkadir, Soeren S. Lienkamp, Thomas Brox, Olaf Ronneberger
   Link: https://arxiv.org/abs/1606.06650

8. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization
   Authors: Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, Dhruv Batra
   Link: https://arxiv.org/abs/1610.02391